Ayush Kumar (Ayush Baranwal)

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Experience

Google Summer of Code Contributor [Project] [Technical Blogs] [SBSCL] Remote
National Resource for Network Biology (NRNB) | Java, C, SBML, JUnit *May 2025 – Present*
- Promoted to **maintainer**; modernizing core via automated **CI/CD** workflows, dependency upgrades, & legacy refactoring.
- Architected a modular, production-ready **LSODA** adaptive ODE solver in **SBSCL**, an open-source systems biology simulation library, outperforming existing solvers in runtime via automatic stiff/nonstiff method switching..
- Built a comprehensive JUnit-based testing pipeline, with unit, intergration, and system tests, validating numeric accuracy and stability against the **1800+ model SBML Test Suite** with a verified **98%+ benchmark pass rate**.
- **Co-first author** of the BioML Symposium 2026 poster *"Simulating Biological Systems with OptSolvX-powered SBSCL"*.

Open Source Developer [PRs] Remote
mpmath (1.2k ⋆) | Python, Numerical Methods *Oct 2025 – Present*
- Developed arbitrary-precision **FFT/IFFT** algorithms (**Cooley-Tukey** and **Bluestein**) in Python, improving support for arbitrary-length Discrete Fourier Transforms within core numerical routines.
- Implemented high-precision **Brent** and **Modified Anderson-Björck** root-finding algorithms, ensuring fast convergence.
- Extended matrix APIs to support list inputs and strengthened edge-case handling, improving usability and robustness.

Artificial Intelligence/Machine Learning Trainee Patna, India
National Institute of Electronics and Information Technology | Python, Scikit-learn, TensorFlow *Jul 2023 – Aug 2023*
- Completed intensive 150+ hour AI/ML training program, achieving **Grade A** certification after final evaluation.
- Built and evaluated 5+ machine learning models covering classification, regression, clustering, DNNs with hands-on experience in data analysis, feature engineering, and hyperparameter tuning.

Selected Projects

Semi-Supervised Liver Segmentation via Hybrid Vision Mamba — *PyTorch, Vim, ViT, MSD Task03*
- Built a **2.5D** pipeline fusing a **CNN + Vision Mamba** encoder with a **TransUNet** decoder for long-range slice context.
- Mitigated edge distortion and foreground imbalance using a **weighted compound loss** (Dice, CE, Focal Tversky, BDoU).
- Designed an automated **pseudo-label reliability scorer** for semi-supervised fine-tuning, achieving a **0.9321 Dice** score.

NumLA - A Scientific Computing Library — *C++, Template Programming* [GitHub]
- Developing a beginner-friendly modern C++ scientific computing library featuring a custom templated matrix engine.
- Implemented linear algebra routines, including Gaussian elimination, LU decomposition, and linear system solvers.

Football Match Analysis System — *Python, Ultralytics, OpenCV, Scikit-learn,, Roboflow, Numpy*
- Built a football match analytics pipeline using **YOLOv8** to detect, track, and analyze players, referees, and the ball.
- Implemented automated team classification, perspective transformation, and Optical Flow-based camera motion estimation.
- Developed algorithms for ball interpolation, possession tracking and estimation of player speed and distance covered.

Achievements

- **All India Rank 466** in **GATE DA (Data Science & Artificial Intelligence) 2026** among over 69,000 candidates.
- **Institute Topper** at **International Collegiate Programming Contest (ICPC) 2025** preliminary contest.
- **Winner** at **EngQuest 3.0** (Intra-School Techfest, SoE JNU) & **ClubFest 2025**.
- **Finalist and Special Mention** at DSEU's **Prod-E-G Hackathon 2024**.
- **Competitive Programming:** Pupil (Max. 1317) *@Codeforces*, 3-Star (Max. 1755) *@Codechef*, 500+ *@LeetCode*.

Education

Jawaharlal Nehru University, New Delhi, India
Bachelor of Technology in Electronics and Communication Engineering — ***CGPA: 8.01/9.0*** *2023 – 2027*

New Government Polytechnic, Patna, India
Diploma in Computer Science and Engineering — ***CGPA: 9.13/10.0 (2022 Institute Rank 1)*** *2020 – 2023*

Technical Skills

- **Programming Languages:** Python, C++, C
- **Frameworks & Libraries:** TensorFlow, Pytorch, Scikit-learn, Pandas, NumPy, OpenCV, Langchain, HuggingFace
- **Developer Tools:** Git, GitHub, VS Code, Linux, Jupyter Notebook
- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Natural Language Processing, Machine Learning, Computer Vision, Operating System, Linear Algebra, Probability & Statistics

Leadership

- President, Co.L.D. (The Coding and Infosec Club), School of Engineering, JNU (2025-26)
- Student Coordinator, Placement Cell, Jawaharlal Nehru University (2026-27)
- Co-Lead Convenor, HackJNU 4.0, JNU's flagship 36-hour annual national-level hackathon.